

Real-time biosignal acquisition and viewing platform

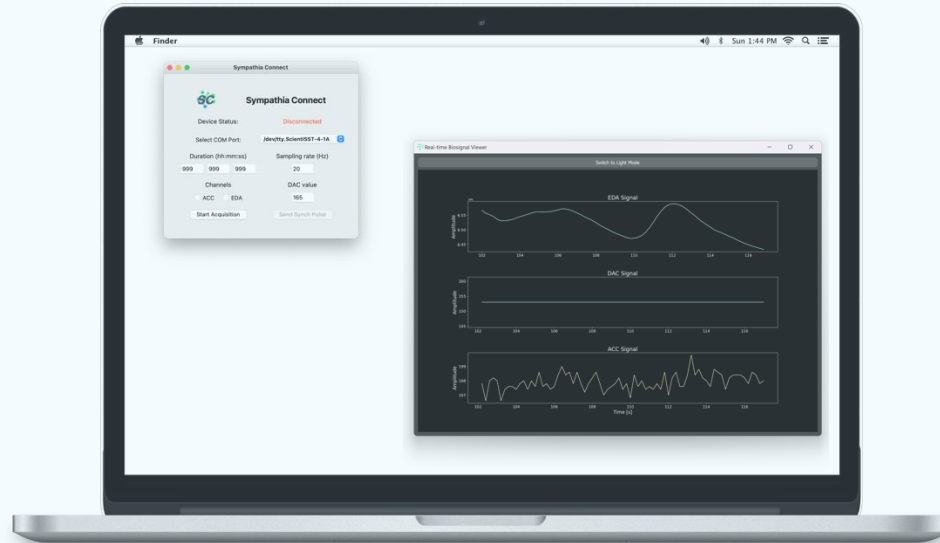
Sympathia Connect User's Guide

Instructions Manual

App version: v1.0.0-beta

User's Guide revision: Initial Release

Sympathia **Connect**



Sympathia **Sense**



Introduction

Sympathia Connect is a UI platform that allows users to collect biosignals from **Sympathia Sense**, a novel smart sock for Electrodermal Activity (EDA) and Acceleration (ACC) monitoring.

Applications

- Research studies in clinical fields (e.g. neurology, psychiatry, sleep disorders)
- Assessment of stress levels
- Assessment of engagement levels and motion sickness in Virtual Reality (VR)
- Arts and Artistic performance
- Neuromarketing

Features

- Connect to Sympathia Sense via **Bluetooth**
- Customized acquisition parameters
 - Channels** – EDA and/or ACC
 - Time** - seconds to hours
 - Sampling rate** – 1 to 8,000 Hz
- Real-time data visualization

System Requirements

The App requires the computer to have a working Bluetooth device, but does not require internet connection to operate (only to update).

- Windows
 - App distribution format: .EXE
 - OS Requirements $\geq$ Windows 7 (with Service Pack 1 or later)
 - Notes:
- MacOS
 - App distribution format: .dmg
 - OS requirements $\geq$ macOS 10.9 (Mavericks) or later

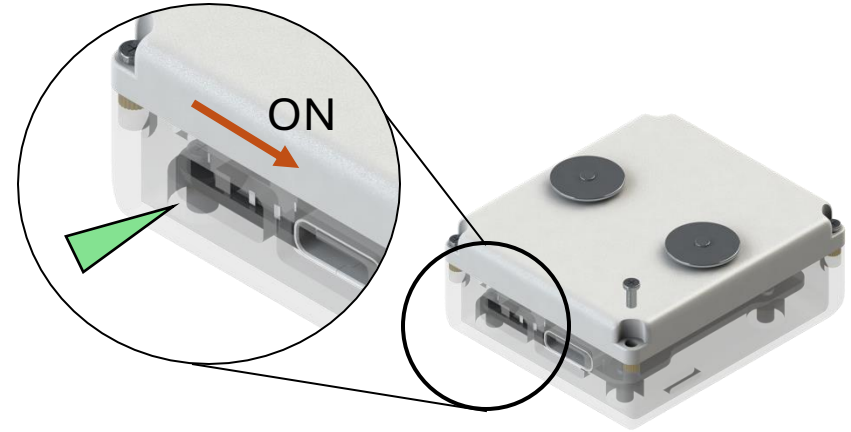
Setup

No installation is required. Move the folder 'Sympathia Connect' to the desired location in your computer, and unzip it if necessary.

Note: Don't remove or change any of the files inside resources subfolder 'rsc'.

Tutorial

1. **Turn the Sympathia Sense device ON.** Use a sharp object to move the switch towards the USB-C connector.
2. **Turn the Bluetooth ON in the computer.**
3. **Connect to device.** The device shows as `ScientISST-xx-xx` (xx-xx being the letters and numbers). If the device does not show, make sure:
 - The device was charged overnight (~8h)
 - Enable **advanced options** in the Bluetooth settings of your computer (the device is not recognized as a standard Bluetooth device).



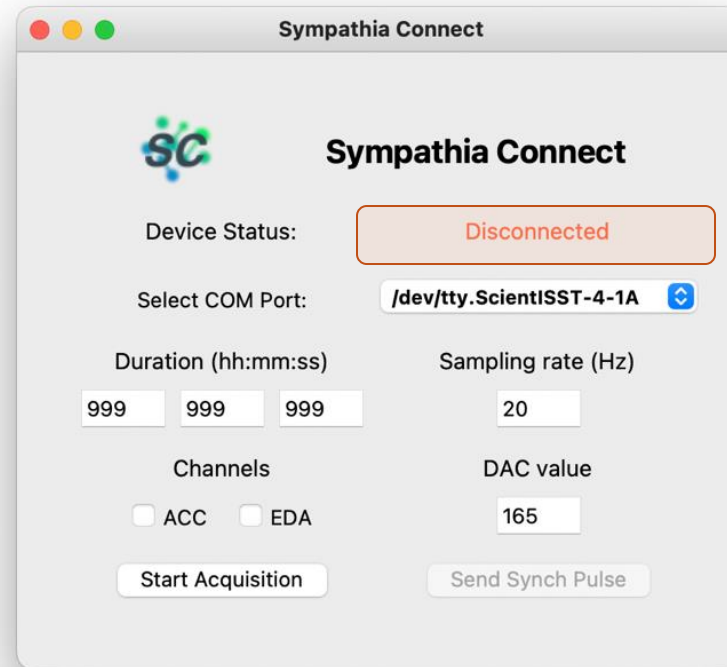
Tutorial

4. **Open the App** by clicking on the file .exe / .dmg (Windows / MacOS).

 rsc	25/11/2024 16:48	Pasta de ficheiros	
 Signals	26/11/2024 18:59	Pasta de ficheiros	
  Sympathia Connect	26/11/2024 19:05	Aplicação	66 047 KB

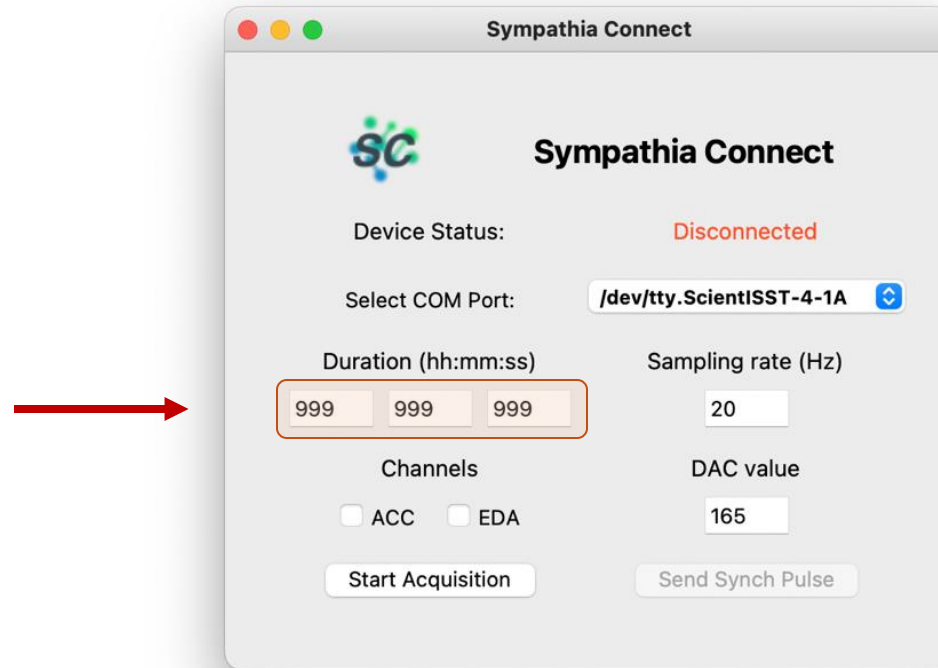
Tutorial

5. **Select the device name** (same as in the Bluetooth options of the computer)



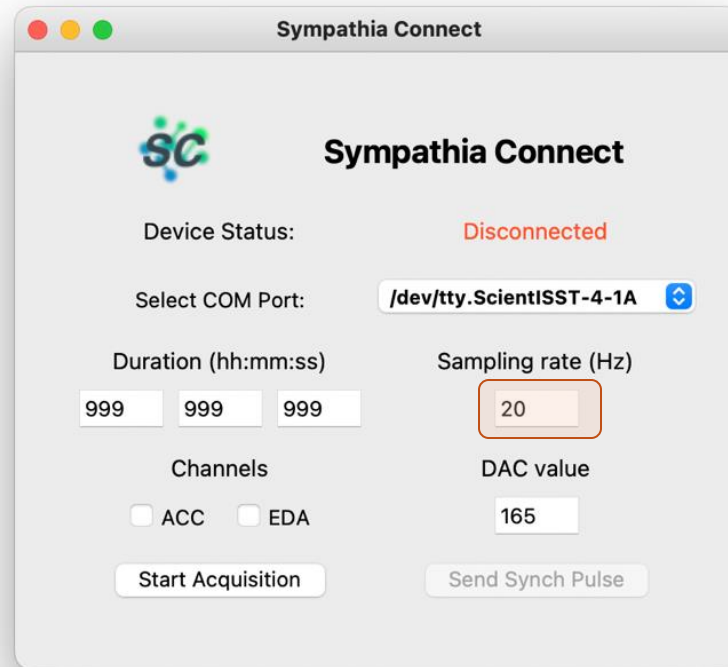
Tutorial

6. Set the acquisition duration in hours, minutes and seconds



Tutorial

7. Set the sampling rate in Hertz (per second)



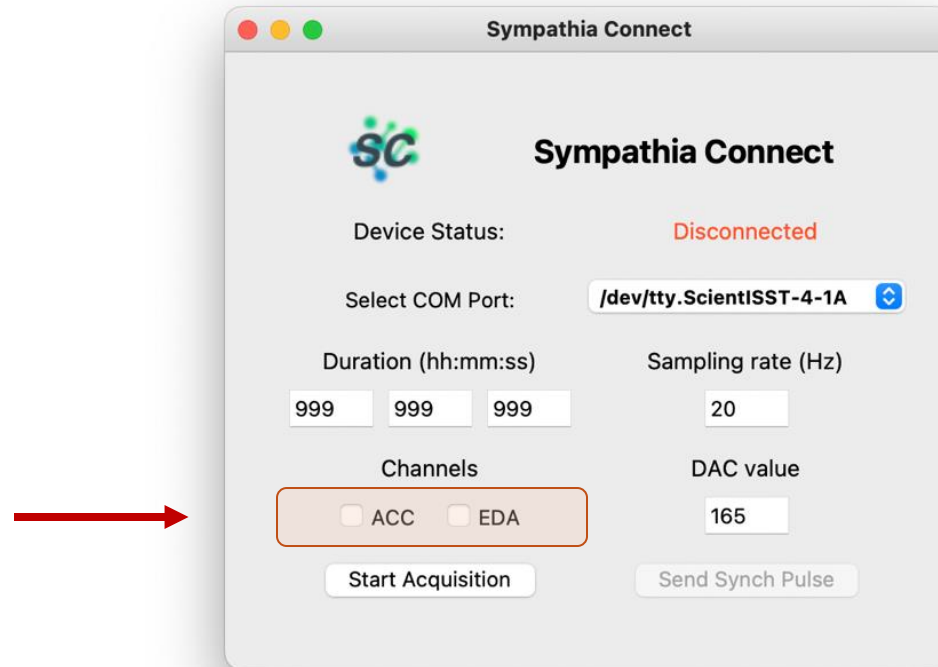
The screenshot shows the Sympathia Connect application window. The title bar reads "Sympathia Connect". Inside the window, there is a logo on the left and the text "Sympathia Connect" on the right. Below the logo, the "Device Status" is "Disconnected". The "Select COM Port" dropdown is set to "/dev/tty.ScientISST-4-1A". The "Duration (hh:mm:ss)" is set to "999 999 999". The "Sampling rate (Hz)" is set to "20", which is highlighted with an orange box and a red arrow. The "Channels" section has checkboxes for "ACC" and "EDA", both of which are unchecked. The "DAC value" is set to "165". At the bottom, there are two buttons: "Start Acquisition" and "Send Synch Pulse".

Sampling rate configurable up to 8,000 Hz.

- **If only EDA is enabled:** > 5 Hz is recommended.
- **If ACC is enabled** (alone or with EDA): > 20 Hz is recommended to capture normal human body movement.

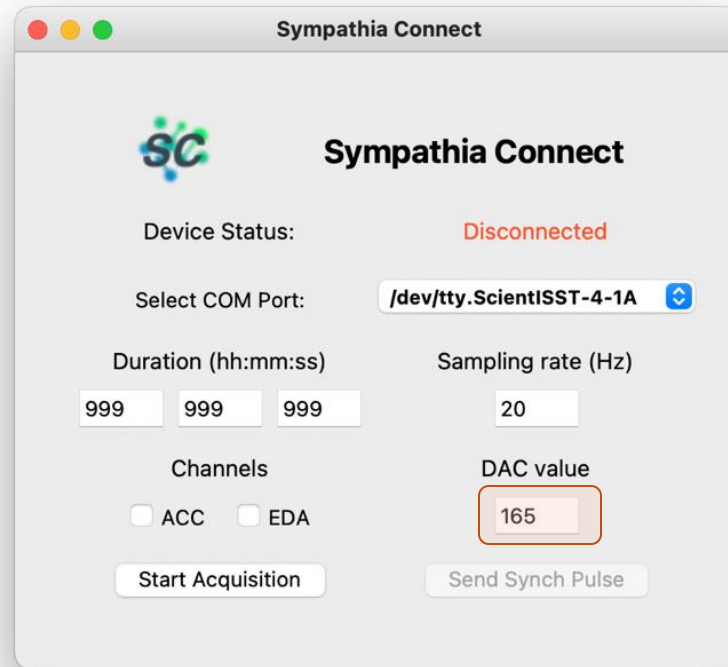
Tutorial

8. Select the biosignal modalities to collect, Acceleration signals and/or Electrodermal Activity



Tutorial

9. Set the starting value for the EDA Sensor configuration, from 0 to 255

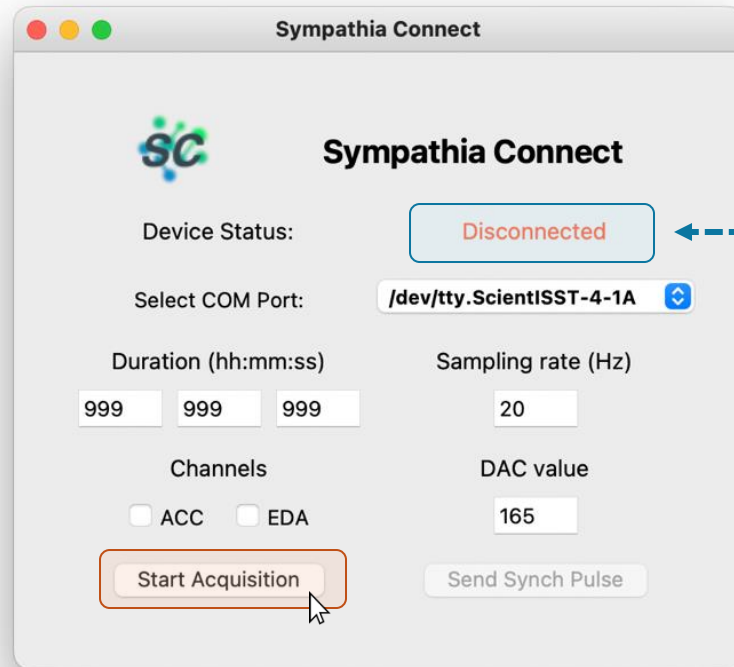


The screenshot shows the Sympathia Connect application window. The title bar reads "Sympathia Connect". Inside the window, there is a logo on the left and the text "Sympathia Connect" on the right. Below the logo, the "Device Status" is "Disconnected". The "Select COM Port" dropdown is set to "/dev/tty.ScientISST-4-1A". The "Duration (hh:mm:ss)" fields are all set to "999". The "Sampling rate (Hz)" is set to "20". Under "Channels", the "ACC" checkbox is checked and the "EDA" checkbox is unchecked. The "DAC value" field is a text input box containing the number "165", which is highlighted with a red rectangular border. A red arrow points from the right side of the window to this "DAC value" field. At the bottom, there are two buttons: "Start Acquisition" and "Send Synch Pulse".

DAC value is the initial value for the sensor 12-bit configuration [0-255]. The value 165 (half-range) is recommended for a first acquisition in a given subject.

Tutorial

10. Start the Acquisition



Note: The **Device Status** displays the status of the currently connected device, and it alternates between: **Disconnected**, **Connecting**, **Connected** and **Acquiring**.

Tutorial

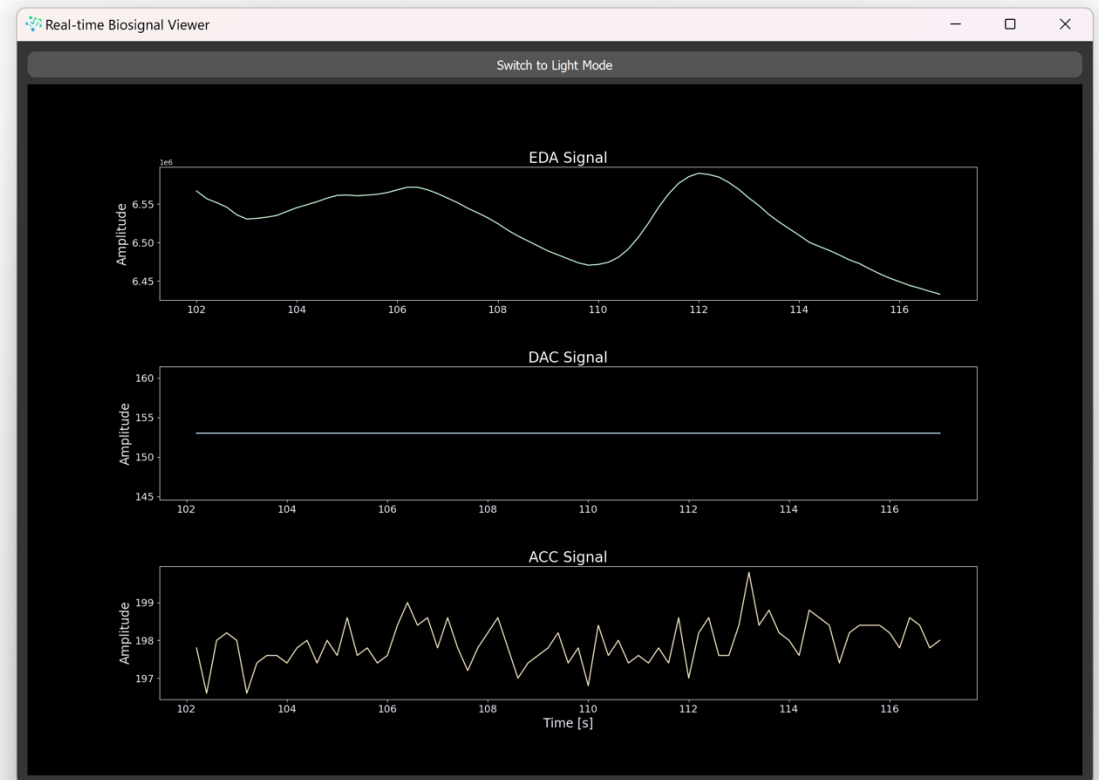
11. Visualize the signals in real-time (separate window)

Note: DAC signals are showed only when the EDA is enabled. It lets the user understand the behavior of the sensor and the trend of the sweat levels (i.e. increasing / decreasing)

EDA

DAC

ACC



Tutorial

11. Visualize the signals in real-time (separate window)

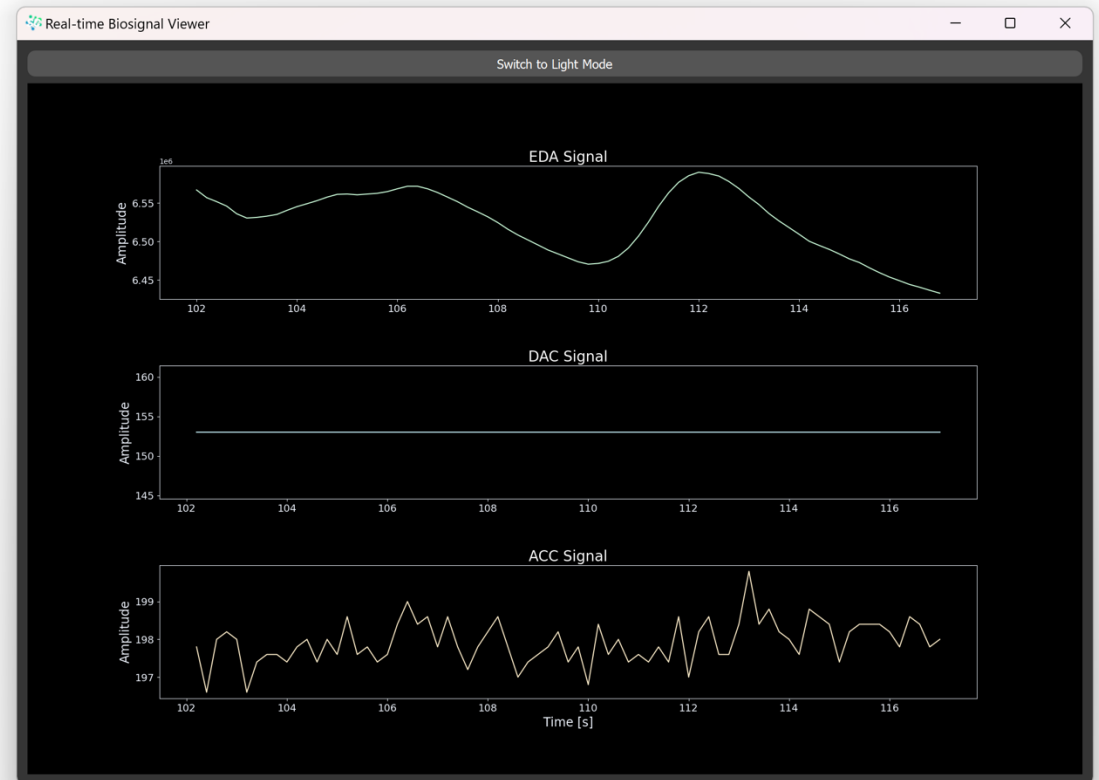
Change between Light and Dark modes



EDA

DAC

ACC



Tutorial

12. The acquisition .csv file is saved in ./Signals

Channels →

↓ Samples

1	#{'API version': '1.1.0', 'Channels': [7],						
2	#NSeq DAC I1 I2 01 02 AX7						
3	0	160	0	0	0	0	0
4	1	160	0	0	0	0	0
5	2	160	0	0	0	0	0
6	3	160	0	0	0	0	0
7	4	159	0	0	0	0	0
8	5	159	0	0	0	0	0
9	6	159	0	0	0	0	0
10	7	159	0	0	0	0	0
11	8	159	0	0	0	0	0
12	9	159	0	0	0	0	0
13	10	159	0	0	0	0	0
14	11	159	0	0	0	0	0
15	12	159	0	0	0	0	0
16	13	159	0	0	0	0	0

Header correspondence:

Signal type	Channel ID in File Header
EDA	AX7
DAC	DAC
ACC	AI1
Ignore others	(...)

Period ΔT (milliseconds) = 1 / Sampling rate (Hz)

ΔT {

13	10	159	0	0	0	0	0
14	11	159	0	0	0	0	0

Time between consecutive samples